ImbutoSmart AI

NISR 2026 Big Data Hackathon

Complete Project Documentation and Technical Overview

Repository: github.com/manziemile80/imbutosmart-ai

Project Vision

From agricultural data to better decisions, from better decisions to better harvests — online or offline.

1. Project Overview

ImbutoSmart AI is an intelligent agricultural decision-support platform designed to transform Rwanda's agricultural and meteorological data into practical, localized and understandable advice for farmers.

The system uses agricultural survey data, district climate information, weather forecasts, machine-learning models and agronomic knowledge to help farmers make better decisions about what to grow, when to act and how to manage their crops.

A Random Forest classifier estimates the probability that a crop will succeed under the selected district, soil and seasonal/weather conditions, while a Random Forest regressor estimates expected yield in tonnes per hectare (t/ha). Crops are ranked according to estimated success and expected yield.

The platform extends these predictions with district climate information, coming-days weather forecasts, soil and crop suitability, fertilizer and manure guidance, seasonal agricultural advice, agricultural risk information, insurance awareness, agronomist assistance, AI-powered questions and answers, voice assistance, multilingual text support, and hybrid online/offline agricultural recommendations.

The system is designed to remain useful whether the farmer has an internet connection or not.

2. Problem Statement

Rwandan farmers need clear and localized information to decide what to plant and how to manage production under different soil, seasonal and weather conditions.

Agricultural, weather and agronomic information can be difficult to combine into one practical decision. Farmers may have access to individual pieces of information but lack a system that brings these sources together and translates them into simple recommendations.

Connectivity is also an important challenge. Some farmers may experience unreliable, expensive or unavailable internet connectivity. A system that depends completely on continuous internet access may therefore become unavailable exactly when a farmer needs agricultural guidance.

ImbutoSmart AI addresses these challenges by combining NISR agricultural survey information with district climate, forecast weather, machine-learning models and a separate agronomic suitability engine. Its hybrid architecture also allows essential agricultural recommendations to continue working when there is no internet connection, using locally stored agricultural information, models and cached data.

3. Main Purpose

The purpose of ImbutoSmart AI is to transform agricultural and meteorological data into simple, localized and actionable advice that helps farmers:

- Choose suitable crops.
- Understand the probability of crop success.
- Estimate potential yield.
- Understand rainfall and seasonal conditions.
- Manage soil and agricultural inputs.
- Understand manure and fertilizer options.
- Decide when to act based on seasonal and weather conditions.
- Understand agricultural risks and make informed insurance decisions.
- Ask agricultural questions using AI.

- Receive assistance in understandable languages.
- Continue receiving essential recommendations even when internet connectivity is unavailable.

4. Core Farmer Questions

- Which crops are suitable for my district and soil?
- What is the estimated probability that each crop will succeed?
- What yield can I expect in tonnes per hectare?
- What are the usual rainfall and temperature conditions in my district?
- What weather is expected in the coming days?
- What agricultural inputs are commonly used in my district?
- What are the observed application rates of these inputs?
- What type of manure or fertilizer may be appropriate?
- When should I act based on seasonal and forecast conditions?
- What insurance should I ask about when production risk is high?
- Can I receive advice in simple language?
- Can I receive information in Kinyarwanda, English or French?
- Can I still receive crop recommendations when there is no internet?

5. Data and Evidence

5.1 NISR Seasonal Agricultural Survey 2025

The training data come from NISR Seasonal Agricultural Survey 2025 microdata covering Seasons A, B and C. The prepared training table contains 51,225 crop observations across 30 districts, 12 crops and 3 seasons. The crop names are mapped to the crop categories offered by the application.

The survey records district, crop, area, harvest information and sampling weight. The microdata and trained models remain outside the public repository and are built locally.

5.2 District Climate and Weather

Because the survey does not record plot-level soil type or plot-level weather, machine-learning training rows use district climate for the relevant season. The project builds district climate using the published 2001–2020 record from NASA POWER and MERRA-2 at district-centre locations, together with weather information from the 2025 survey year.

The farmer interface separates usual district climate from coming-days weather forecast. Historical climate information comes from NASA POWER, while coming-days forecasts come from Open-Meteo.

6. Machine-Learning Models

6.1 Crop Success Classifier

A Random Forest classifier estimates the probability that a crop will succeed under district, season and weather conditions represented by the model.

6.2 Expected Yield Regressor

A Random Forest regressor estimates expected yield in tonnes per hectare. The system can use the expected yield together with crop-success probability to rank potential crops.

6.3 Model Evaluation

On a held-out fifth of the survey data, the current project reports classifier AUC 0.94 and yield-regressor R^2 0.81, with per-crop yield errors including 0.84 t/ha for maize and 0.29 t/ha for beans. The project also notes that 2025 was a below-average rainfall year, meaning that learned yields reflect conditions from a relatively dry survey year.

7. Soil, Manure and Agronomic Advice

Farmer-selected soil is not treated as a measured variable in the machine-learning training model. Instead, soil is used by a separate knowledge-based agronomic suitability engine. This distinction prevents the system from incorrectly implying that the NISR survey directly measured plot-level soil.

The Advice portal provides district-level information about DAP, NPK, urea, manure, compost and other agricultural inputs, including observed application rates in kg/ha, irrigation indicators and soil-erosion indicators. These are aggregate district profiles rather than household-level information.

The system can also provide agronomic guidance explaining how soil characteristics and agricultural inputs relate to crop suitability.

8. Seasonal Rainfall and When to Act

The system combines the district's usual seasonal rainfall and temperature information with coming-days weather forecasts. This helps farmers understand expected seasonal conditions, current weather conditions, potential weather risks, appropriate timing for agricultural activities and when to monitor weather conditions before taking action.

Rainfall information is presented as statistical or forecast guidance. The system does not guarantee that a particular amount of rainfall will occur.

9. Agricultural Insurance Advice

Insurance-related guidance is included in the farmer summary. The system helps farmers identify agricultural insurance products or information worth asking about when production risk is high. Insurance guidance is intended as decision support and awareness. It is not an insurance contract and does not guarantee coverage, eligibility or payment.

10. AI Assistance

The farmer Q&A; assistant and season explanations use an OpenAI-compatible chat-completions endpoint. The farmer assistant can answer agricultural questions using the evidence supplied to it. The agronomist assistant uses figures displayed by the dashboard, model report and district climate information.

The AI is instructed to use supplied evidence and state when a required figure or information is missing instead of inventing information.

- Hosted AI.
- Local AI.
- Farmer Q&A.;
- Agricultural explanations.
- Seasonal explanations.

- Agronomist assistance.

Supported hosted options include Groq. A local AI option uses Ollama with Qwen 2.5 3B.

11. Voice and Language Support

The Guidance portal provides Ask a question, Speak your question, This season and My plan.

- English.
- Kinyarwanda.
- French.

The local speech configuration supports speech-to-text and text-to-speech depending on the configured speech backend.

- Whisper / faster-whisper for speech-to-text.
- Kokoro.
- pyttsx3.
- Other configured text-to-speech backends.
- FFmpeg for audio conversion.

Reliable spoken Kinyarwanda and French support depends on the speech backend configured for the deployment.

12. Hybrid Online and Offline AI Capability

One of the important features of ImbutoSmart AI is its ability to work in both connected and disconnected environments.

The system follows a hybrid architecture:

Connected when possible, useful when offline, synchronized when connected again.

This makes the platform more practical for farmers who may not have reliable internet access.

12.1 Online Mode

When internet connectivity is available, ImbutoSmart AI can use online services to provide updated information.

- Retrieve coming-days weather forecasts.
- Access the shared agronomist database.
- Synchronize farmer and agronomist information.
- Provide AI-powered questions and answers.
- Retrieve updated seasonal information.
- Access updated agronomic records.
- Provide updated input and insurance information where available.
- Synchronize records between devices.
- Use online voice services where configured.

The online mode provides the most current information available to the system.

12.2 Offline Mode

When internet connectivity is unavailable, ImbutoSmart AI continues providing essential agricultural decision support using locally stored information and models.

- Crop recommendations.
- Soil-based crop suitability.
- Estimated crop-success probability.
- Expected yield where the required model is available.
- Seasonal agricultural guidance.
- Stored district climate information.
- Soil and agronomic guidance.
- Previously downloaded manure and fertilizer information.
- Previously synchronized insurance information.
- Local AI assistance where the local model is installed.
- Previously downloaded agricultural knowledge.

This means a farmer can still receive useful crop recommendations without an active internet connection.

12.3 Offline AI

For AI assistance without internet access, the application can use a locally installed AI model. The documented local architecture supports Ollama with Qwen 2.5 3B. The local model can provide agricultural explanations and answer suitable farming questions using the information available locally.

The local AI must use the available evidence and should clearly state when information is unavailable. It should never invent weather, prices, insurance rules or agricultural statistics.

12.4 Online and Offline Synchronization

The system uses synchronization to keep locally stored information updated.

The farmer opens ImbutoSmart AI.

The system checks network availability.

If internet is available, current information is retrieved.

Important information is cached locally.

If the internet connection is lost, the application switches to locally available information.

The farmer continues receiving agricultural recommendations.

When the connection returns, the system synchronizes updated information.

Outdated cached information can then be replaced with newer information.

The system should display the current status clearly: **Online**, **Offline**, or **Synchronizing**. This helps the farmer understand whether information is live or stored locally.

13. Application Structure

The project uses FastAPI, Python, scikit-learn, React, TypeScript/TSX, Supabase, machine-learning models, weather services, optional AI and voice services, and local/offline storage and models.

The backend contains data preparation, machine-learning pipelines and API code. The frontend contains the Farmer Advisory and Agronomist Dashboard interfaces.

```
backend/ FastAPI + scikit-learn
src/ React web application
src/components/ui/ Reusable UI components
src/styles.css Shared styles and themes
```

14. Agronomist Dashboard

The /dashboard page is the advisory workspace for agricultural officers. It contains modules for overview, manure and fertilizer, agronomist contacts, farmer profiles, crop-advice review, soil and nutrients, weather risk, pests and disease, insurance, farmer messages, knowledge centre, reports and follow-up, and offline data.

The dashboard reads the same district, soil and season selected by the farmer. Published figures come from NISR survey aggregates, district climate records, live Open-Meteo forecasts and trained-model reports.

The dashboard does not estimate a missing figure. A price, premium, application rate, eligibility rule or risk score is displayed only when an agronomist records a verified value with its supplier or policy, source and verification date. A price older than 90 days is labelled for re-checking.

15. Shared Agronomist Data

Editable records are stored in a shared Supabase project. This allows authorized devices to access the same information. The application includes database migrations for agronomist records, audit records, reviewer allowlists and row-level security.

The shared database is the primary path when configured. When the connection is unavailable, the system can maintain an offline copy on the device and synchronize changes when connectivity returns.

16. Offline Data Management

The offline data system is particularly important for agricultural field visits. Agronomists can use previously synchronized information during visits where network connectivity is unavailable.

- Farmer profiles.
- Crop recommendations.
- Agricultural knowledge.
- Input information.
- Manure information.
- Agronomic guidance.
- Insurance information.
- Previously synchronized climate information.

When connectivity returns, the system can synchronize applicable records with the shared cloud database.

17. Privacy and Data Protection

- Survey microdata are not committed to the public repository.
- Trained model artifacts are protected.
- District input profiles contain aggregate information.
- District land-use imports are aggregate and non-sensitive.
- AI credentials remain server-side.

- Email credentials are stored in environment variables.
- Private farmer information is not unnecessarily exposed.

18. Technologies Used

Frontend Technologies

- React.
- TypeScript / TSX.
- CSS.
- Reusable React UI components.

Backend Technologies

- Python.
- FastAPI.
- Uvicorn.
- scikit-learn.
- Joblib.

Data and Machine Learning

- NISR Seasonal Agricultural Survey 2025.
- NASA POWER.
- MERRA-2.
- Open-Meteo.
- Pandas.
- Python data-processing workflows.

AI and Voice

- OpenAI-compatible Chat Completions API.
- Groq.
- Ollama.
- Qwen 2.5 3B.
- Whisper / faster-whisper.
- Kokoro.
- pyttsx3.
- FFmpeg.

Database and Deployment

- Supabase.
- PostgreSQL.
- Row-Level Security.

• Render.

• Vercel.

• Netlify.

• Docker.

• Environment variables.

19. Recommendation Flow

The farmer selects a district.

The farmer selects their soil type.

The farmer selects the agricultural season.

The application checks network availability.

If online, the application obtains current district climate and forecast information.

If offline, the application uses locally stored climate and agricultural information.

Machine-learning models evaluate crop success probability.

The yield model estimates potential yield.

The agronomic engine evaluates soil and crop suitability.

District agricultural input information is added.

Manure and fertilizer guidance is provided.

Seasonal and weather information helps determine when to act.

Agricultural risk information is considered.

Insurance awareness is included where relevant.

The AI assistant explains the recommendation.

The system produces a simple farmer-facing agricultural plan.

When the internet returns, updated information is synchronized.

20. Example Recommendation

Information	Example
District	Gatsibo
Soil	Farmer-selected soil type
Season	Season A
Weather	District climate + coming-days forecast
Connectivity	Online or Offline
Crop ranking	Crops ranked by estimated success and expected yield
Yield	Displayed in t/ha
Input advice	District-level observed input use + agronomic guidance
Risk advice	Seasonal/weather risk + insurance guidance

The recommendation can provide the farmer with an understandable plan instead of requiring the farmer to interpret raw agricultural datasets.

21. Running the Project

Backend

```
cd backend
pip install -r requirements.txt
python prepare_training_data.py --microdata "C:/path/to/Microdata"
python ml_pipeline.py
uvicorn main:app --reload --port 8000
```

Frontend

```
cp .env.example .env
npm install
npm run dev
```

For production, the frontend and backend can be deployed using the documented hosting architecture.

22. Deployment Architecture

The documented deployment architecture uses Render for the FastAPI backend and Vercel, Netlify or Render for the frontend. The training data and models are prepared privately and the required trained model artifacts are supplied to the backend.

After deployment, the system should verify /meta, a real /predict request and /health.

The hybrid architecture also requires local/offline data and model support so that essential farmer recommendations remain available without continuous internet access.

23. Advantages of ImbutoSmart AI

- Data-driven recommendations based on agricultural survey data, climate information and machine-learning models.
- Localized advice based on district, soil and season.
- Yield estimation in tonnes per hectare.
- Climate awareness using historical and forecast information.
- Agronomic guidance for soil, manure, fertilizers and agricultural inputs.
- Insurance awareness for agricultural risk.
- AI assistance for agricultural questions.
- Voice assistance through configured speech technologies.
- Multilingual written support in English, Kinyarwanda and French.
- Offline capability for essential recommendations.
- Synchronization when connectivity returns.
- Agronomist support for reviewing farmer information and recommendations.

24. Limitations

The system has several limitations that should be communicated clearly.

- Offline mode cannot provide live information unless it was previously synchronized.
- Coming-days weather forecasts may become outdated while offline.
- Newly published agricultural information cannot be retrieved offline.
- Updated input prices cannot be retrieved without connectivity.
- New insurance information cannot be retrieved without connectivity.
- Cloud-based agronomist records cannot synchronize until connectivity returns.
- Local AI performance depends on the device's available hardware.
- Local speech support depends on the installed speech backend.

The system should therefore indicate whether information is live, previously synchronized, historical, model-generated or agronomic guidance.

25. Future Development

- More detailed soil datasets.
- More real-time meteorological sources.
- Reliable Kinyarwanda speech-to-text.
- Reliable Kinyarwanda text-to-speech.
- Crop disease and pest advisory.
- Market-price information.
- Market-access information.
- Deeper agricultural-insurance integration.
- Mobile application development.
- SMS delivery for farmers.
- Improved offline AI models.
- More extensive offline agricultural datasets.
- Automatic synchronization conflict management.

26. NST2 Alignment

ImbutoSmart AI supports NST2 priorities related to agricultural productivity, improved crop selection, higher potential yield per hectare, climate-resilient agriculture, data-driven agricultural decision-making, agricultural extension services and digital transformation.

The hybrid architecture additionally supports accessibility for farmers who experience limited connectivity.

27. AI Disclosure

AI assistants were used to help architect the machine-learning pipeline, generate UI components and debug code. The core data strategy, NISR data-processing logic and system design were implemented by the project team.

AI-generated recommendations are intended to support agricultural decision-making and should not be interpreted as guaranteed agricultural outcomes.

28. Project Vision

ImbutoSmart AI aims to bridge the gap between Rwanda's agricultural and meteorological data and the everyday decisions made by farmers. The platform transforms complex data into practical, localized and understandable agricultural guidance.

Its hybrid architecture ensures that agricultural decision support is not completely dependent on internet connectivity.

The long-term vision is to create a reliable AI agricultural assistant for Rwanda that combines agricultural statistics, weather information, climate data, machine learning, soil suitability, agronomic knowledge, fertilizer and manure information, agricultural insurance awareness, voice assistance, local AI, offline technology, online services and agronomist expertise.

Connected when possible, useful when offline, synchronized when connected again.

“From agricultural data to better decisions, from better decisions to better harvests — online or offline.”

ImbutoSmart AI is therefore not only a crop recommendation system. It is a hybrid AI-powered agricultural decision-support platform designed to help Rwandan farmers make better, data-driven and climate-aware farming decisions regardless of whether they currently have internet connectivity.